

# 2026-06-10-SP2-sdk-bump-plan

## SP2 — SDK Bump Plan (provider/infra SDK currency)

**Revision:** 1 **Created:** 2026-06-10 **Last modified:** 2026-06-10 **Maintainer:** CLI-Agent Fusion programme (SP2 P2.3 SDK-currency) **Status:** active **Status summary:** Read-only planning artefact — ordered, risk-classified go.mod bump sequence for HelixCode’s provider/infra SDKs.

READ-ONLY PLAN. This document changes nothing. It cites the currency research at docs/research/2026-06-10-sdk-cli-currency.md (PART A) and the exact import-site evidence gathered from the two go.mod files + the \*.go import statements. Every bump step below is to be executed one SDK at a time per §11.4.42 iteration discipline, §11.4.92 multi-pass, §11.4.40 full-suite-retest, and §11.4.108 four-layer verification.

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### 1. Inventory — current pins vs latest

Module IDs are both dev.helix.code (inner) / dev.helix.agent (helix\_agent submodule). Inner app go.mod = helix\_code/go.mod; submodule = submodules/helix\_agent/go.mod.

| SDK module                               | helix_code pin                  | helix_agent pin | LATEST (research §A) | Skew     | Same           |
|------------------------------------------|---------------------------------|-----------------|----------------------|----------|----------------|
| github.com/aws/aws-sdk-go-v2 (core)      | v1.32.7 (helix_code/go.mod:16)  | — (not used)    | v1.42.0              | 10 minor | yes (v1.42.0)  |
| github.com/aws/aws-sdk-go-v2/config      | v1.28.7 (helix_code/go.mod:17)  | —               | siblings of core     | —        | yes (v1.28.7)  |
| github.com/aws/aws-sdk-go-v2/credentials | v1.17.48 (helix_code/go.mod:18) | —               | siblings of core     | —        | yes (v1.17.48) |

| SDK module                                                      | helix_code pin                                                   | helix_agent pin                                           | LATEST (research §A)                                                                                             | Skew                                      | Same                                |
|-----------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------|
| github.com/<br>aws/aws-sdk-<br>go-v2/service/<br>bedrockruntime | v1.23.1 (helix_code/<br>go.mod:19)                               | —                                                         | <b>v1.53.5</b>                                                                                                   | <b>~30<br/>minor</b>                      | yes (v                              |
| github.com/<br>aws/smithy-go                                    | v1.22.1 (helix_code/<br>go.mod:20)                               | —                                                         | (transitive<br>of AWS<br>core)                                                                                   | —                                         | yes                                 |
| github.com/<br>Azure/azure-<br>sdk-for-go/<br>sdk/azcore        | v1.16.0 (helix_code/<br>go.mod:13)                               | —                                                         | v1.22.0                                                                                                          | 6 minor                                   | yes (v                              |
| github.com/<br>Azure/azure-<br>sdk-for-go/<br>sdk/azidentity    | v1.8.0 (helix_code/<br>go.mod:14)                                | —                                                         | <b>v1.13.1</b>                                                                                                   | 5 minor                                   | yes (v<br><b>SEC</b><br><b>GO-2</b> |
| github.com/<br>getzep/zep-go/<br>v3                             | v3.10.0 (helix_code/<br>go.mod:27)                               | —                                                         | v3.23.0                                                                                                          | 13<br>minor                               | yes (v                              |
| github.com/<br>gin-gonic/gin                                    | <b>v1.12.0</b> (helix_code/<br>go.mod:28)                        | v1.12.0<br>(submodules/<br>helix_agent/<br>go.mod:5)      | v1.12.0                                                                                                          | <b>none<br/>—<br/>already<br/>aligned</b> | yes                                 |
| github.com/<br>smacker/go-<br>tree-sitter                       | v0.0.0-20240827094217-<br>dd81d9e9be82<br>(helix_code/go.mod:41) | same commit<br>(submodules/<br>helix_agent/<br>go.mod:73) | smacker at<br>HEAD<br>(dormant);<br>maintained<br>alt =<br>tree-<br>sitter/<br>go-<br>tree-<br>sitter<br>v0.25.0 | at<br>HEAD                                | n/a —<br>migrat<br>bump             |

## Important fact-correction vs the research’s pin snapshot

The research (docs/research/2026-06-10-sdk-cli-currency.md:13-16) recorded helix\_code gin as **v1.11.0**. The CURRENT helix\_code/go.mod:28 already reads gin v1.12.0 — the gin skew described in the research summary item 6 **is already resolved** (per the gitStatus this session: “gin already fixed to 1.12”). Both modules are on gin v1.12.0. **No gin bump remains.** This is captured here so the executor does not re-do already-landed work (§11.4.37 fetch-before-edit / §11.4.114 known-good).

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## 2. Classification

| Bump                                                           | Class                                                                       | Rationale                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| service/<br>bedrockruntime<br>v1.23.1 → v1.53.5                | <b>BREAKING-RISK</b><br>(large jump, needs its own careful sub-task)        | Same v1 major, but ~30 minor jump. Generated AWS service clients add model/operation/types churn across 30 minors; the import surface in <code>bedrock_provider.go</code> includes <code>.../bedrockruntime/types + ResponseStreamReader + InvokeModelWithResponseStream</code> — event-stream + types are the historically churning areas. Treat as code-change-possible, isolate. Core+config+credentials+smithy <b>MUST</b> move together (shared internal versions).                                                                   |
| aws-sdk-go-v2<br>core+config+credentials+smithy-go             | <b>SAFE-COORDINATED</b><br>(must move as a set, before/with bedrockruntime) | Same v1 line. v1.42.0's new standard-retry is opt-in behind <code>AWS_NEW_RETRIES_2026</code> env (non-breaking per research §A row 1). These four + the ~10 AWS indirect modules ( <code>internal/configsources</code> , <code>internal/endpoints/v2</code> , <code>service/internal/presigned-url</code> , <code>sso</code> , <code>ssooidc</code> , <code>sts</code> , <code>protocol/eventstream</code> , etc. at <code>helix_code/go.mod:82-91</code> ) share lock-step internal versions; <code>go mod tidy</code> resolves the set. |
| azidentity v1.8.0 → v1.13.1                                    | <b>SECURITY</b><br>(prioritise — GO-2024-2918 EoP)                          | Same v1 line. Carries the Azure Identity elevation-of-privilege advisory on old releases (research §A row 4). Stay on <b>stable v1.13.1</b> ; v1.14.0 is beta-only — do NOT adopt beta.                                                                                                                                                                                                                                                                                                                                                    |
| azcore v1.16.0 → v1.22.0                                       | <b>SAFE</b>                                                                 | Same v1 line, additive (research §A row 5). azidentity v1.13.1 likely pulls a newer azcore as a minimum anyway, so doing azcore alongside/just-before azidentity avoids a double tidy.                                                                                                                                                                                                                                                                                                                                                     |
| getzep/zep-go/<br>v3 v3.10.0 → v3.23.0                         | <b>SAFE</b>                                                                 | Same v3 line, additive (research §A row 2). Single importer file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| gin                                                            | <b>DONE / no-op</b>                                                         | Already v1.12.0 in both modules. No action.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| smacker/go-tree-sitter → tree-sitter/go-tree-sitter<br>v0.25.0 | <b>DEFER</b> (separate migration epic, NOT in this bump batch)              | Different import path + different API (Close-on-allocation), 30+ language sub-package imports across <code>internal/repomap/tree_sitter.go</code> , <code>internal/repomap/tag_extractor.go</code> , <code>internal/tools/mapping/treesitter_parsers.go</code>                                                                                                                                                                                                                                                                             |

| Bump | Class | Rationale                                                                                                                                                                                                                       |
|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |       | (helix_code) + submodules/<br>helix_agent/internal/clis/<br>aider/repo_map.go (helix_agent).<br>No vendor supersession statement<br>(research §A migration note). Real<br>migration work — do NOT fold into a<br>currency bump. |

### 3. Per-bump detail (edit + importers + verification + rollback)

For every step: pre-flight `git fetch --all --prune (§11.4.37)`, branch off `main` (never edit `go.mod` on `main` directly), run the bump, then the listed verification. Rollback for ALL = `git checkout -- go.mod go.sum && go mod tidy` (or `git stash/git restore` to the pre-bump SHA). All commands run from `helix_code/` unless noted.

#### 3.1 — getzep/zep-go/v3 (SAFE, do first — smallest blast radius)

- **Edit:** `helix_code/go.mod:27 v3.10.0 → v3.23.0`.
- **Importer (only one prod file):** `internal/memory/providers/zep_provider.go` — imports `getzep/zep-go/v3 (zep)`, `.../v3/client (zepclient)`, `.../v3/option (zep_provider.go:14-16)`. 1342 lines; the API surface to re-verify is the `zep client` constructor + memory `get/add` option chains.
- **Command:**  
  

```
go get github.com/getzep/zep-go/v3@v3.23.0 && go mod tidy &&
    go build ./... \
    && go test ./internal/memory/...
```
- **§11.4.40 verification:** `go build ./... clean + go vet ./internal/memory/... + go test -count=1 ./internal/memory/...` GREEN. If a real-Zep integration test exists (`-tags=integration`), run it against the live endpoint per CONST-050(A) (no mocks beyond unit).
- **Rollback:** `git checkout -- go.mod go.sum && go mod tidy`.

#### 3.2 — azcore (SAFE) then azidentity (SECURITY) — do as a pair

- **Edit:** `helix_code/go.mod:13 azcore v1.16.0 → v1.22.0`; `helix_code/go.mod:14 azidentity v1.8.0 → v1.13.1`. Also the indirect `sdk/internal v1.10.0 (helix_code/go.mod:78)` will likely bump via `tidy`.
- **Importers:** `internal/llm/azure_provider.go` — imports `sdk/azcore`, `sdk/azcore/policy`, `sdk/azidentity (azure_provider.go:18-20)`; uses `azcore.TokenCredential`, `azidentity.NewManagedIdentityCredential`, `azidentity.ClientID`, `azidentity.NewDefaultAzureCredential`

(azure\_provider.go:39,150,252,261-270). Test: internal/llm/azure\_provider\_test.go.

- **Command:**

```
go get github.com/Azure/azure-sdk-for-go/sdk/azcore@v1.22.0
go get github.com/Azure/azure-sdk-for-go/sdk/azidentity@v1.13.1
go mod tidy && go build ./... && go test ./internal/llm/...
```

- **§11.4.40 verification:** build clean + go test -count=1 ./internal/llm/... GREEN. The credential constructors above are the contract to re-verify (managed-identity + default-cred signatures are the most likely churn points). SECURITY note: confirm go.sum now records azidentity ≥ v1.13.1 (the GO-2024-2918-patched line) — capture go list -m github.com/Azure/azure-sdk-for-go/sdk/azidentity output as the §11.4.5 evidence the advisory is cleared. Optionally run govulncheck ./internal/llm/... to prove GO-2024-2918 no longer reports.

- **Rollback:** git checkout -- go.mod go.sum && go mod tidy.

### 3.3 — aws-sdk-go-v2 core set (SAFE-COORDINATED) — gate for bedrockruntime

- **Edit:** helix\_code/go.mod:16 core v1.32.7 → v1.42.0; :17 config; :18 credentials; :20 smithy-go — bump together; let go mod tidy settle the ~10 AWS indirect modules at helix\_code/go.mod:82-91.

- **Importers:** internal/llm/bedrock\_provider.go — aws-sdk-go-v2/aws, .../config(awsconfig), .../credentials, .../smithy-go (bedrock\_provider.go:13-15,18); uses smithy.APIError (bedrock\_provider.go:1038), config.LoadDefaultConfig-style NewFromConfig plumbing.

- **Command:**

```
go get github.com/aws/aws-sdk-go-v2@v1.42.0 \
      github.com/aws/aws-sdk-go-v2/config@latest \
      github.com/aws/aws-sdk-go-v2/credentials@latest \
      github.com/aws/smithy-go@latest
go mod tidy && go build ./... && go test ./internal/llm/...
```

(Resolve config/credentials to the versions tidy picks as compatible with core v1.42.0 rather than pinning a literal — they track core.)

- **§11.4.40 verification:** build clean + go test -count=1 ./internal/llm/... GREEN. Do NOT set AWS\_NEW\_RETRIES\_2026 (leave retry behaviour unchanged — research §A row 1). Capture go list -m github.com/aws/aws-sdk-go-v2 as evidence.

- **Rollback:** git checkout -- go.mod go.sum && go mod tidy.

### 3.4 — service/bedrockruntime (BREAKING-RISK — own careful sub-task)

Do this ONLY after 3.3 lands GREEN (the core set is the dependency floor for the service client).

- **Edit:** `helix_code/go.mod:19 v1.23.1 → v1.53.5`.
- **Importers:** `internal/llm/bedrock_provider.go` — `service/bedrockruntime + service/bedrockruntime/types` (`bedrock_provider.go:16-17`). Concrete API surface to re-verify (the churn-risk points across 30 minors):
  - `InvokeModel(ctx, *bedrockruntime.InvokeModelInput, ...func(*bedrockruntime.Options) InvokeModelWithResponseStream(...)` interface methods (`bedrock_provider.go:24-25`);
  - `bedrockruntime.NewFromConfig(awsCfg)` (`bedrock_provider.go:238`);
  - `InvokeModelInput / InvokeModelWithResponseStreamInput` struct fields (`bedrock_provider.go:456,493`);
  - `bedrockruntime.ResponseStreamReader` event-stream iface in `processEventStream` (`bedrock_provider.go:880`) — **highest churn risk** (event-stream types move most).
  - Tests: `internal/llm/bedrock_provider_test.go`, `internal/llm/response_err_round54_test.go`.
- **Command:**

```
go get github.com/aws/aws-sdk-go-v2/service/bedrockruntime@v1.53.5
go mod tidy && go build ./... && go test ./internal/llm/...
```
- **If `go build ./...` FAILS** (the breaking-risk case): the failure is the §11.4.108 ARTIFACT/SOURCE signal. Per §11.4.102 systematic-debugging FIRST — read the compile error, diff the changed types/`ResponseStreamReader/*Input` symbol against v1.53.5's GoDoc (verify against `pkg.go.dev/github.com/aws/aws-sdk-go-v2/service/bedrockruntime` per §11.4.99 latest-source), update the call sites in `bedrock_provider.go` minimally, re-build. This is the one bump where code edits are expected, so it is split into its own sub-task and its own commit.
- **§11.4.40 verification:** `build clean + go test -count=1 ./internal/llm/...` GREEN +, if a real-Bedrock integration test gate exists, a live `InvokeModel` round-trip producing a real response (anti-bluff per §11.9 — NEVER a simulated response per BLUFF-001). Capture the real AWS response as §11.4.69 `network_connectivity / LLM-output` evidence.
- **Rollback:** `git checkout -- go.mod go.sum internal/llm/bedrock_provider.go && go mod tidy`.

### 3.5 — smack → tree-sitter migration (DEFER — not part of this batch)

- **Why deferred:** different import path (github.com/tree-sitter/go-tree-sitter) + different API (mandatory Close() on allocations). Importers to migrate (count = real work):
    - helix\_code: internal/repomap/tree\_sitter.go (sitter + 9 language subpkgs, tree\_sitter.go:9-18), internal/repomap/tag\_extractor.go(:8), internal/tools/mapping/treesitter\_parsers.go (sitter + **30 language subpkgs**, treesitter\_parsers.go:6-33), internal/tools/mapping/doc.go, plus test internal/repomap/incremental\_p2t06\_test.go;
    - helix\_agent: submodules/helix\_agent/internal/clis/aider/ repo\_map.go.
  - **Blocker:** the 30+ smack language sub-packages (/bash, /c, /cpp, /golang, ...) do NOT have a 1:1 mapping in tree-sitter/go-tree-sitter — each language is a separate tree-sitter-<lang>/bindings/go module. This is a multi-file, multi-module migration epic that MUST be its own plan with its own §11.4.43 RED tests per language. **Track as a separate workable item; do not bump in the currency batch.** smack is at HEAD, so currency is not regressing by leaving it.
- 

## 4. Ordered execution plan

One SDK (or coordinated set) per branch/commit, each fully verified before the next. All in helix\_code (helix\_agent has none of these except the deferred tree-sitter — see §5).

1. **zep-go/v3** (SAFE, smallest) — §3.1. Commit on GREEN.
2. **azcore + azidentity** (SAFE + SECURITY pair) — §3.2. **Prioritised within the safe tier because of GO-2024-2918.** Capture the patched-version evidence. Commit on GREEN.
3. **aws core set** (core+config+credentials+smithy, SAFE-COORDINATED) — §3.3. Commit on GREEN. This is the dependency floor for step 4.
4. **service/bedrockruntime** (BREAKING-RISK, own sub-task) — §3.4. Expect possible code edits in bedrock\_provider.go; systematic-debugging on any compile break; own commit.
5. **(DEFER)** smack → tree-sitter migration — §3.5. NOT in this batch; spin a separate plan.

After EACH step: go build ./... + the affected-package go test -count=1 GREEN before proceeding (§11.4.42). After the WHOLE batch: full go test -count=1 ./... + go vet ./... + anti-bluff smoke (§9 of root CLAUDE.md) + §11.4.125 code-review gate before any tag.

Ordering rationale: safe→security→breaking, BUT the AWS core set (3.3) must precede bedrockruntime (3.4) because the service client depends on the core/internal modules' versions; within the “safe” tier azidentity is pulled forward to clear the security advisory early.

## 5. helix\_code vs helix\_agent coordination

- **bedrockruntime, aws core set, azcore, azidentity, zep-go/v3 — helix\_code ONLY.** submodules/helix\_agent/go.mod does NOT require any of these (confirmed: it has no aws-sdk-go-v2, no Azure/azure-sdk-for-go, no getzep/zep-go). No cross-module coordination needed for steps 1-4.
- **gin** — both modules **already on v1.12.0** (helix\_code/go.mod:28, submodules/helix\_agent/go.mod:5). The “gin-skew” the research warned of is already closed;

no action. (Keep them aligned in future bumps — gin-skew style coordination still applies if either moves.)

- **smacker/go-tree-sitter** — pinned at the SAME commit in BOTH modules (helix\_code/go.mod:41, submodules/helix\_agent/go.mod:73). The deferred migration is the ONE item needing gin-skew-style cross-module coordination: when it runs, helix\_code's 5 files
  - helix\_agent's repo\_map.go must migrate together so the pinned commit / new module stays consistent across both go.mod files. Plan that as a coordinated two-module epic, not a bump.

## 6. Final ordered bump list + code-change verdict

| # | SDK                                                                                        | go.mod line(s)                                                         | Class                     | Code changes?                                                                                                                                                        |
|---|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | getzep/zep-go/<br>v3<br>v3.10.0→v3.23.0                                                    | helix_code/<br>go.mod:27                                               | SAFE                      | <b>No</b> (pure go.mod; verify zep_provider.go compiles)                                                                                                             |
| 2 | azcore<br>v1.16.0→v1.22.0 +<br>azidentity<br>v1.8.0→v1.13.1                                | helix_code/<br>go.mod:13,14                                            | SAFE +<br><b>SECURITY</b> | <b>No</b> expected (pure go.mod; verify azure_provider.go creditors)                                                                                                 |
| 3 | aws core: aws-<br>sdk-go-v2<br>v1.32.7→v1.42.0 +<br>config +<br>credentials +<br>smithy-go | helix_code/<br>go.mod:16,17,18,20                                      | SAFE-<br>COORDINATED      | <b>No</b> expected (pure go.mod; tidy settles ~10 indirects)                                                                                                         |
| 4 | service/<br>bedrockruntime<br>v1.23.1→v1.53.5                                              | helix_code/<br>go.mod:19                                               | <b>BREAKING-<br/>RISK</b> | <b>Possibly YES</b> —<br>bedrock_provider.go<br>types/<br>ResponseStreamReader/<br>*Input may need edits; own<br>sub-task + own commit                               |
| 5 | smacker→tree-<br>sitter/go-<br>tree-sitter<br>v0.25.0                                      | helix_code/<br>go.mod:41 +<br>submodules/<br>helix_agent/<br>go.mod:73 | <b>DEFER</b>              | <b>YES (heavy)</b> — import-path<br>+ API migration across 5<br>helix_code files + 1<br>helix_agent file + 30+<br>language subpkgs; separate<br>epic, NOT this batch |

**Pure-go.mod (no code expected):** steps 1, 2, 3. **Code-change candidate (its own careful sub-task):** step 4 (bedrockruntime 30-minor jump). **Out of scope / separate epic (definite code migration, two-module coordination):** step 5 (tree-sitter import-path migration).